WhyCode

Why is this code like this? Ask its own history.

Team FORTEC · hackFront India 2026 · AI and Developer Tools (T-1)

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Built on django/django

Nobody can explain the code anymore

- Codebases accumulate decisions nobody can explain any more.
- A new developer reads an odd function and guesses why it exists.
- A wrong guess breaks something the original author fixed on purpose.
- The reasons live scattered across commits, pull requests and tickets — never in one place, never linked to the code they explain.

WhyCode: ask the history, get the evidence

- Ingest a public repo’s history — commits, PRs and tickets — into a graph.
- Ask plain-language “why” questions about the code.
- Get a cited evidence chain plus a confidence score.
- When the graph holds no evidence, it refuses. No invented answers, ever.

From repo history to cited answers

1 · Ingest

GitHub API via gh (commits, PRs) plus Trac ticket pages — Django tracks decisions in Trac, not GitHub Issues.

2 · Graph

Entities: commit, PR, ticket, file, person. Edges: authored, fixes, references, touches, supersedes.

3 · Ask

Match the question, walk the evidence chain 2 hops, flag decisions superseded by newer ones — and refuse when evidence is absent.

4 · Runs anywhere

The bundled SQLite database needs no credentials and no paid APIs.

A real answer, with cited evidence

```
$ python -m whycode.cli --question "Why does ignore_warnings match substrings by default?"
refused: false    confidence: 0.99
Best match: Fixed #37089 — Made ignore_warnings() match substrings by default. (commit)
Evidence:
→ fixes      Trac ticket #37089
→ authored by HelmiDev03
```

Same for the template-tags option

```
$ python -m whycode.cli --question "Why is there a transitional option to disable multiline template tags?"
refused: false    confidence: 0.86
Best match: Refs #35899 — Added allow_multiline_tags transitional template option. (commit)
Evidence:
→ references Trac ticket #35899
→ authored by boxed
```

No evidence? It refuses instead of inventing

```
$ python -m whycode.cli --question "Why does Django use a microservices architecture?"
refused: true    confidence: 0.0
No evidence in the graph answers this. Refusing to invent an answer.
```

An earlier version answered this from an unrelated docs commit — the absent-term rule + regression test fixed it for good.

Runnable, tested, no magic

200 commits	**100** pull requests	**100** Trac tickets
161 files	**90** people	**630** edges

- Python standard library only · SQLite graph · gh CLI.
- 10/10 automated tests, including refusal and supersession cases.
- Bundled database runs the whole demo with zero credentials.

WhyCode — by FORTEC

github.com/eres0045/hackfront-fortec

```
$ python -m whycode.cli --db data/whycode.db --question "Why ...?"
```

Questions welcome.

Built during the hackFront league timeline, from Django’s real, public history.